

# EB-JEPA for Clinical EEG

Self-Supervised Abnormality Detection: Honest Results

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# Why Self-Supervised EEG?

## Clinical reality:

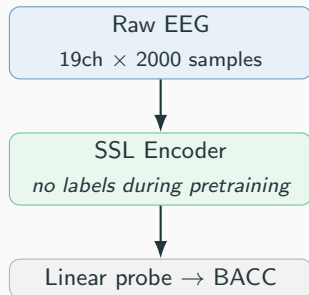
- Every EEG must be reviewed by a licensed neurologist
- Labels are scarce and expensive
- Unlabelled EEG data is abundant

**Goal:** learn a *transferable* EEG representation **without labels**

**Hypothesis:** corruption augmentation + weak invariance ( $\lambda=1$ ) is the dominant lever, not encoder scale.

## Benchmark: TUAB v3.0.1

- Binary: normal / abnormal clinical EEG
- 19 channels, 200 Hz, 10 s windows
- 2717 train / 276 eval (patient-disjoint)



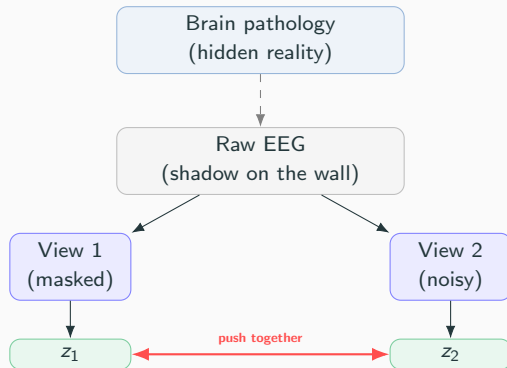
# Plato's Cave: What SSL Actually Does

Prisoners in Plato's cave see only **shadows on the wall**, not the real objects. They learn to *predict* patterns in the shadows without ever seeing the cause.

## In our setting:

- Cave wall = raw EEG voltages
- Hidden reality = underlying brain pathology
- SSL encoder = prisoner learning that *two corrupted views of the same shadow are the same object*
- Linear probe = asking the prisoner to name what they see

The encoder never sees a neurologist's label. It learns structure from the signal alone.



# EB-JEPA: Two-View Invariance + Corruption

## Two-view SSL pipeline:

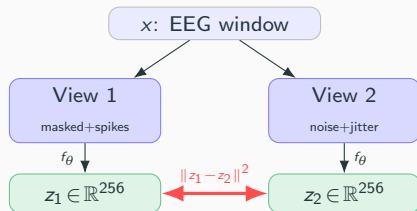
1. Take one EEG window; create two views  $v_1, v_2$
2. Encode:  $z_i = f_\theta(v_i) \in \mathbb{R}^{256}$
3. Minimise  $\|z_1 - z_2\|^2 + \text{anti-collapse term}$

**Standard augmentations:** noise, jitter, channel drop

## Our design: Corruption augmentation

- 20 % time steps **masked** to zero
- 20 % time steps injected with  $\pm 6\sigma$  **spikes**
- Mimics real EEG artefacts; forces robust learning

**Objectives:** VICReg ( $V+C$ ) and SIGReg (Epps-Pulley)



## Finding 1: Corruption Augmentation as Dominant Lever

| Variant                           | BACC <sub>REC</sub>                | $\Delta$      |
|-----------------------------------|------------------------------------|---------------|
| Base VICReg (no corruption)       | 0.796                              | –             |
| + <b>Corruption</b> (mask+spikes) | <b>0.819 <math>\pm</math> .004</b> | <b>+0.023</b> |
| + Spectral loss                   | 0.795–0.813                        | $\approx 0$   |
| Scaled encoder                    | 0.805                              | –0.014        |
| 4 $\times$ more data              | 0.812                              | –0.007        |

### 3-seed average (robust):

0.819  $\pm$  0.004 (not a lucky seed)

### Why corruption helps:

- Standard augmentation creates views that are “too easy” to match
- Masking + spike injection forces learning *content*, not artefact patterns
- Mirrors real EEG noise distribution

### What does NOT help:

- More encoder capacity  $\rightarrow$  same plateau
  - 4 $\times$  pretraining data  $\rightarrow$  no gain
- $\Rightarrow$  Bottleneck is *architecture*, not scale

## Finding 2 (Novel): EEG Prefers Weak VICReg Invariance

**VICReg loss:**

$$\mathcal{L} = \lambda_{\text{inv}} \underbrace{\|z_1 - z_2\|^2}_{\text{inv}} + 25 V + 1 C$$

Paper-correct:  $\lambda_{\text{inv}} = 25$ . Default:  $\lambda_{\text{inv}} = 1$ .

| $\lambda_{\text{inv}}$ | BACC <sub>REC</sub> | Seeds |
|------------------------|---------------------|-------|
| 1 (default)            | <b>0.819 ± .004</b> | 3     |
| 5                      | 0.820               | 1     |
| 10                     | 0.814               | 1     |
| 25 (paper)             | 0.789 ± .005        | 3     |

Gap  $\lambda=1$  vs  $\lambda=25$ : 3 pp,  $\approx 6\sigma$ , not noise

**Why?**

- With  $\lambda=1$ : anti-collapse ( $V+C$ ) **dominates**
- Encoder spreads representations: more discriminative
- With  $\lambda=25$ : encoder collapses toward view-similarity

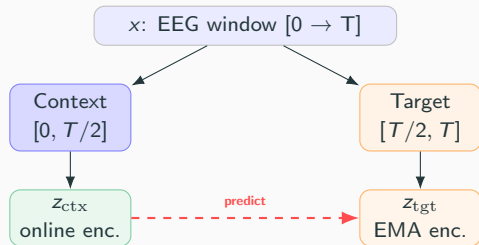
**Sweep**  $\lambda \in \{1, 5, 10, 25\}$ : Sweet spot at  $\{1, 5\}$ ; gap widens monotonically.

First evidence that EEG SSL requires weaker view-invariance than vision SSL.

# NaiveJEPA: Temporal-Split Prediction, No Anti-Collapse Needed

**Idea:** split one window in time instead of augmenting.

- Context ( $0 \rightarrow T/2$ ): online encoder  $\rightarrow$  mean-pool
- Target ( $T/2 \rightarrow T$ ): EMA encoder  $\rightarrow$  mean-pool
- Loss:  $\text{MSE}(\hat{z}_{\text{tgt}}, z_{\text{tgt}})$
- **No VICReg needed:** collapse resisted by BatchNorm (zero variance  $\rightarrow$  explosion) + EMA lag (non-stationary target)



0.04 BACC below VICReg+corruption, with 1 hyperparameter.

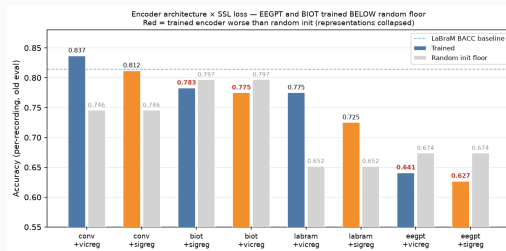
| Method              | BACC <sub>REC</sub> | Hypers   |
|---------------------|---------------------|----------|
| VICReg + corruption | $0.819 \pm .004$    | 4        |
| <b>NaiveJEPA</b>    | <b>0.779</b>        | <b>1</b> |

# Encoder Ranking: Simple Conv Wins

| Encoder       | Type         | Params | BACC                 |
|---------------|--------------|--------|----------------------|
| <b>Conv1D</b> | Strided conv | 0.4M   | <b>0.819</b>         |
| BIOT-style    | FFT patches  | 1.2M   | 0.790                |
| LaBraM-style  | Raw patches  | 1.2M   | $\sim 0.775^\dagger$ |
| <b>EEGPT</b>  | Ch-pool Tr   | 0.8M   | <b>collapsed</b>     |

$^\dagger$ accuracy, not BACC (early eval)

**EEGPT failure:** cross-attention pools 19 ch  $\rightarrow$  1 token; channel-drop creates incompatible views. Fix: remove channel-drop  $\rightarrow$  TUEV  
0.182  $\rightarrow$  0.358.



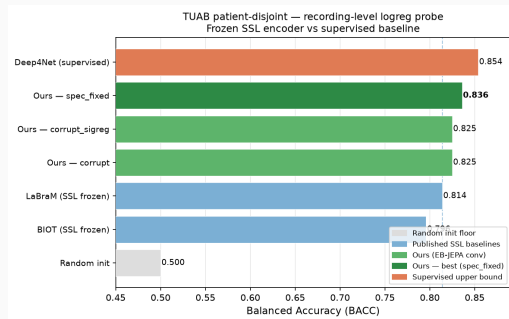
# Honest Position vs Literature

**Evaluation trap:** literature (BIOT, LaBraM) reports **per-window** BACC. Our setup mean-pools 16 windows/patient → **per-recording** (+5 pp for same encoder). Direct comparison manufactures a false win.

| Method                        | BACC <sub>WIN</sub> | AUROC        |
|-------------------------------|---------------------|--------------|
| ContraWR (SSL)                | 0.775               | 0.846        |
| BIOT pretrained               | 0.802               | 0.874        |
| LaBraM-Base                   | 0.814               | 0.902        |
| <b>EB-JEPA SIGReg+corrupt</b> | <b>0.775</b>        | <b>0.856</b> |
| EB-JEPA VICReg+corrupt        | 0.770               | 0.848        |
| EEGNet (supervised)           | 0.796               | 0.882        |

✓ Matches ContraWR (1.6M) with 0.4M params

✗ 3.9 pp below LaBraM (5.8M, 200 ep)



## Cross-Task Transfer to TUEV (6-class Event Detection)

**Protocol:** freeze TUAB-pretrained encoder; train linear probe on TUEV (zero TUEV labels during SSL).

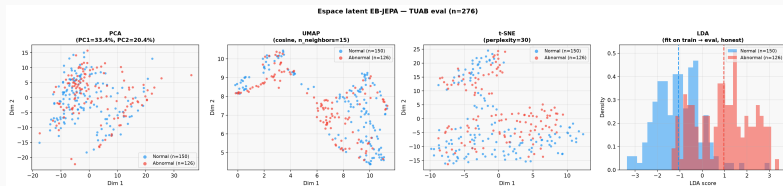
| Method                   | TUEV BACC    | TUEV labels? |
|--------------------------|--------------|--------------|
| ST-Transformer           | 0.3385       | full train   |
| CNN-Transformer          | 0.4100       | full train   |
| <b>VICReg (TUAB SSL)</b> | <b>0.382</b> | probe only   |
| <b>NaiveJEPA (TUAB)</b>  | <b>0.389</b> | probe only   |
| FFCL                     | 0.4551       | full train   |
| ContraWR                 | 0.4664       | full train   |
| BIOT pretrained          | 0.5009       | full train   |

Zero TUEV labels seen during SSL pretraining. A linear probe on frozen features beats **ST-Transformer** and **CNN-Transformer** trained end-to-end on TUEV.

Transfer from a binary task to a 6-class task with only logistic regression: strongest evidence of genuine generality.

We trail BIOT/ContraWR/FFCL because they fine-tune or train from scratch on TUEV.

# Latent Space: EB-JEPA Representations



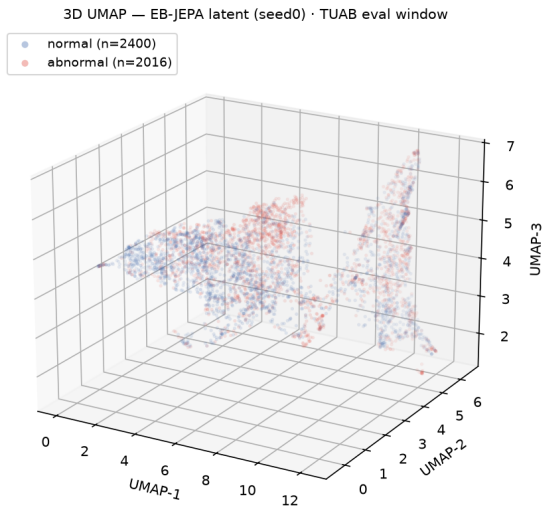
**Setup:** Conv1D +  
VICReg  $\lambda=1$  +  
corruption (BACC  
0.819). 276 eval  
patients, 256-dim.

**PCA/UMAP/t-SNE**  
look mixed. VICReg  
spreads signal across all  
256 dims equally; 2D  
projections lose the  
discriminative direction.  
Expected, not a failure.

**LDA (4th):** fit on *train*  
(2717), projected on  
*eval* (honest).

Near-clean separation  
confirms BACC 0.819 is

# Latent Space: 3D UMAP (EB-JEPA, Window-Level)



**Setup:** same encoder, window-level latents (seed 0), 3D UMAP instead of 2D.

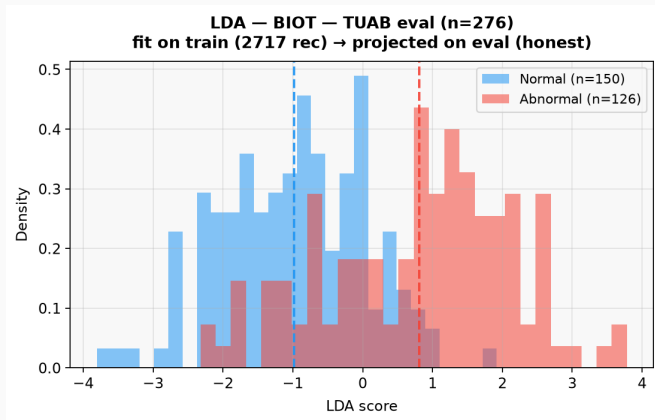
2400 normal / 2016 abnormal windows. The third dimension recovers a two-lobe structure that 2D UMAP collapses.

Still unsupervised projection (no labels used to fit UMAP) — qualitative support for Finding 1, not the BACC evidence (that's LDA, previous slide).

# Latent Space: BIOT (LDA, Same Protocol)

**BIOT pretrained:** EEG-SHHS+PREST-18ch

BACC<sub>WIN</sub> 0.802 (sleep+pathology pretraining, 1.2M)



**Same LDA protocol** as EB-JEPA (slide before last): fit on TUAB *train* (2717 rec), projected on *eval* (276 rec, honest).

EB-JEPA: BACC<sub>REC</sub> 0.819, 0.4M params.

BIOT pretrained on sleep EEG (SHHS); domain mismatch with clinical pathology may reduce separability.

# Summary

## What works

- **Corruption:** +2.3 pp BACC (3-seed, robust)
- $\lambda_{\text{inv}}=1$ : beats  $\lambda=25$  by 3 pp/ $6\sigma$ , EEG-specific
- **SIGReg**  $\geq$  **VICReg**: consistent +0.5–1 pp AUROC
- **NaiveJEPA**: 0.779 BACC with 1 hyperparameter
- **TUEV transfer**: beats 2 supervised baselines, zero TUEV labels

## Open next step

$\lambda_{\text{inv}}$  sweep  $\{1, 5, 10, 25\}$  confirms monotone ordering: EEG SSL prefers weaker invariance, first systematic evidence in the literature.

## What doesn't

- Scaling the encoder (capacity is not the limit)
- $4\times$  pretraining data (conv saturates)
- EEGPT (channel-drop  $\times$  channel-pool conflict)
- Spectral auxiliary loss (neutral)
- GRU predictive JEPA (same-window degenerates)

## Backup: Per-Recording Table (Clinical Metric)

Per-recording = mean-pool 16 windows/patient. Not directly comparable to per-window literature. Best per-window: 0.775 (SIGReg+corrupt).

## Backup: VICReg $\lambda_{\text{inv}}$ Ablation